Hand Derivation: Interbank Conservation and Relative Coverage

1. Interbank market

$$P = \sum_K \mu_K \tau_K x_K, \quad Q = \sum_K \mu_K \tau_K D_K$$

With $P, Q > 0$

$$\psi_K = \tau_K \min\{1, P/Q\}$$
$$\ell_K = \tau_K \min\{1, Q/P\}$$

2. Borrowed liquidity

$$\sum_K \mu_K \psi_K D_K = \sum_K \mu_K \tau_K \min\{1, P/Q\} D_K$$
$$= \min\{1, P/Q\} \sum_K \mu_K \tau_K D_K$$
$$= \min\{1, P/Q\} Q$$
$$= \min\{P, Q\}$$

3. Lent liquidity

$$\sum_K \mu_K \ell_K x_K = \sum_K \mu_K \tau_K \min\{1, Q/P\} x_K$$
$$= \min\{1, Q/P\} P$$
$$= \min\{P, Q\}$$

$$\boxed{\sum_K \mu_K \psi_K D_K = \sum_K \mu_K \ell_K x_K = \min\{P, Q\}}$$

Interbank trade redistributes existing liquidity without creating additional liquidity

If $P = 0$ or $Q = 0$, the transaction volume is defined as zero

4 Common-fund feasibility

$$\Delta = \sum_K \mu_K (1 - \psi_K) D_K$$

$$\phi = \min\{1, F/\Delta\}$$

Then for $\Delta > 0$

$$\sum_K \mu_K \phi (1 - \psi_K) D_K = \phi \Delta$$
$$= \min\{1, F/\Delta\} \Delta$$
$$= \min\{\Delta, F\}$$

Therefore

$$\sum_K \mu_K \phi (1 - \psi_K) D_K \le F$$

When $\Delta = 0$, used coverage is zero

5. Relative coverage incidence

Define

$$C_K = E_A[\phi(A)(1 - \psi_K(A)) D_K(A)]$$
$$\bar{C} = \sum_K \mu_K C_K, \quad D = \sum_K \mu_K D_K$$

If $\bar{C} > 0$

$$I_K = \frac{C_K / d_K}{\bar{C} / D}$$

$$\frac{\mu_K d_K}{D} I_K = \frac{\mu_K C_K}{\bar{C}}$$

$$\sum \frac{\mu_K d_K}{D} I_K = \sum \frac{\mu_K C_K}{\bar{C}}$$

$$= \frac{\bar{C}}{\bar{C}}$$

$$= 1$$

$$\boxed{\sum \frac{\mu_K d_K}{D} I_K = 1}$$

$$B = \{A : \Delta(A) > 0\}, \quad P_B = \Pr(\Delta > 0)$$

If all states with residual deficit still rationalized

$$\bar{C}(F) = F P_B$$

States with zero residual demand contribute zero coverage

6. Fixed-buffer invariance

If buffers and market programs still fixed and relevant states are rationalized

$$C_K = F E_B \left[\frac{(1 - \psi_K) D_K}{\Delta} \right]$$

(in states where $\Delta > 0$)

Therefore

$$I_K = \frac{E_B[(1 - \psi_K) D_K / \Delta] / d_K}{(\sum_J \mu_J E_B[(1 - \psi_J) D_J / \Delta]) / D}$$

With fixed buffers and unchanged rationing regimes, higher common capacity scales absolute coverage but leaves relative coverage invariant.

This does not establish an equilibrium incidence or welfare result.